COMP343 / COMP536 - Digital Forensics

Continuous Assessment 1 (CA1) Report



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1. Executive Summary

This forensic report details a digital forensics examination of the M57 Jean data set using Autopsy 4.20. The purpose of this investigation was to determine how sensitive employee data was created, released, and exposed by an employee where that information was stored; in this case, by Jean. File system metadata analysis revealed that the confidential data was stored in a spreadsheet file called m57biz.xls, which was created on Jean's computer on July 20th, 2008. Examination of email artifacts showed that Jean had received an urgent request for employee information that included names, salary, and social security numbers. Believing this to be a legitimate request, Jean complied and sent the requested information back to the author via email (in an attachment). This information was sent to an external email address and later made available online resulting in a confirmed data breach to multiple employees. The incident was based on a social engineering attack that used urgency and trust.

2. Objectives and Scope of the Investigation

To determine how confidential company data was revealed, a systematic digital forensics investigation was done of a disk image taken from the computer of Jean, Chief Financial Officer of M57.biz.

The investigation aimed to address the following key questions:

1. **When was the spreadsheet file m57biz.xls created?**
2. **How was the spreadsheet exfiltrated from Jean's computer and published on an external website?**
3. **Whether other individuals within the organisation were involved in the incident.**

3. Evidence Analysed

The investigation examined a range of digital artefacts extracted from the forensic disk image of Jean's computer. The primary evidence sources included:

3.1 File System Artefacts

The Windows file system contained a spreadsheet file which is named m57biz.xls, located within Jean's user profile on the desktop. Metadata associated with this file provided timestamps indicating creation, modification, and access times.

3.2 Email Communications

A Microsoft Outlook PST archive recovered from the disk image contained extensive email correspondence between Jean and other employees. Several messages were identified as directly relevant to the incident, including:

- An email requesting sensitive employee data
- A reply from Jean indicating that the requested information had been attached
- Follow up messages discussing unusual events
- Communications reporting that employee personal data had been exposed online

3.3 Email Attachments

The attached files to the appropriate emails were reviewed to determine whether the provided spreadsheets originated outside the company. The results support that the data required was sent as an attachment within the provided spreadsheet shown in Jean's records.

4. Tools and Methods Employed

The forensic examination utilized Autopsy 4.20, an open-source digital forensic tool. The forensic disk image of Jean’s computer (nps-2008-jean.E01) was used as a primary data source and analysed in a restricted virtual environment to maintain evidence integrity.

The following Autopsy ingest modules were enabled during analysis:

- **Recent Activity:** Identified recently accessed files and user activity indicators.
- **File Type Identification:** Detected document types and enabled location of spreadsheet files.
- **Email Parser:** Extracted messages and attachments from Outlook PST archives.
- **Embedded File Extractor:** Recovered files contained within email attachments.
- **Keyword Search:** Enabled targeted keyword searches.
- **Interesting Files Identifier:** Flagged potentially significant documents for further examination.
- **Extension Mismatch Detector:** Identified files whose extensions did not match their actual type.

5. Findings and Analysis

5.1 Creation of the Spreadsheet m57biz.xls

Metadata analysis shows that this file was created on July 20th, 2008, at 01:28:03 UTC. This Excel spreadsheet has sensitive information about employees including their names, salaries, and social security numbers. The timing of the file's creation is relevant because it corresponds to subsequent emails requesting this data. This is evidence that the data in this file was created on Jean's computer; it wasn't downloaded from an external source. Also, the timestamps on the Outlook PST file show that there are PST file activities, they do not correspond to the creation time on the attached Excel spreadsheets.

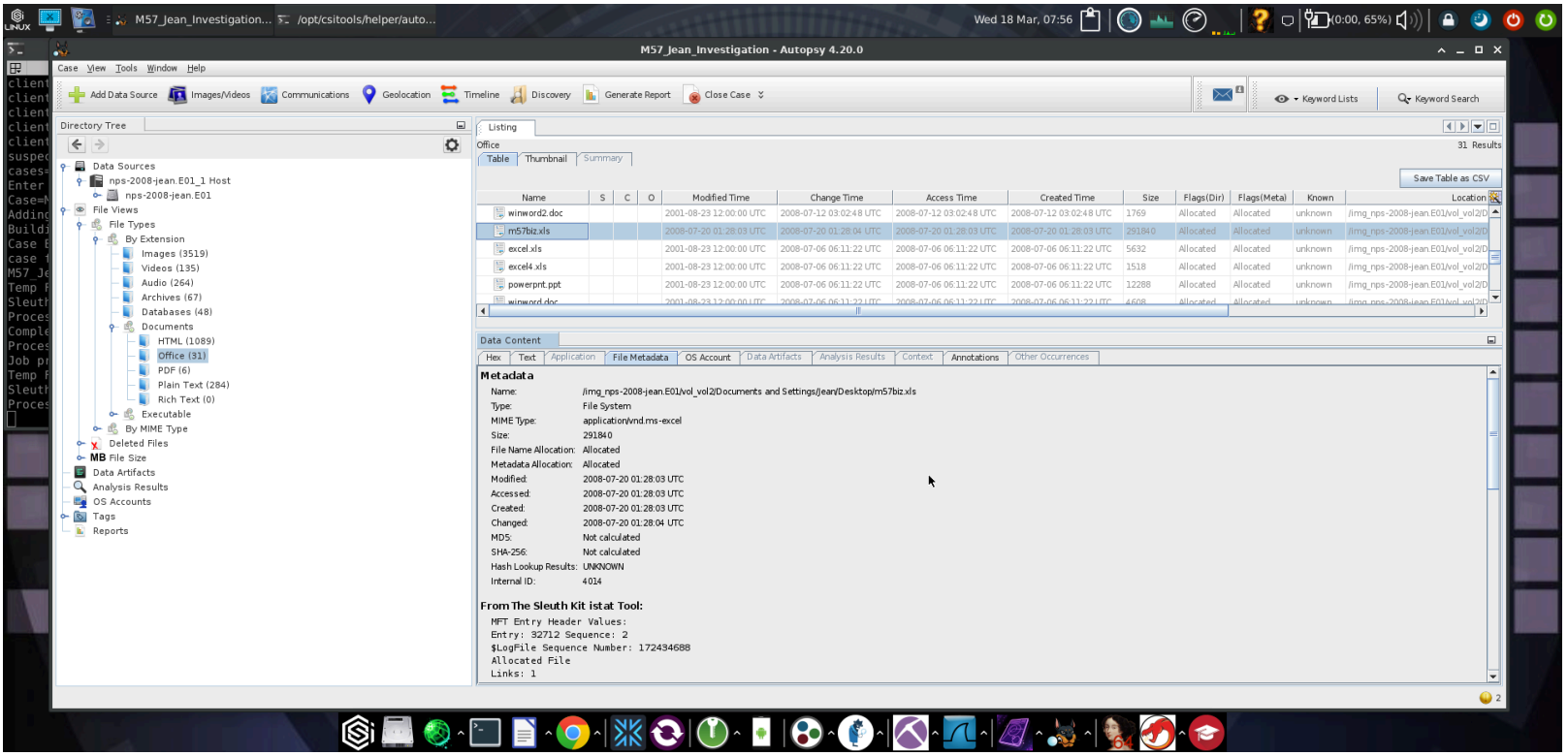


Figure 1: Metadata showing creation time and location of m57biz.xls on Jean's Desktop

5.2 Suspicious Request for Confidential Information

Analysis of the email archive revealed that Jean received an urgent message requesting sensitive employee information, including names, salaries, and Social Security Numbers. The message claimed that the information was required immediately for business purposes.

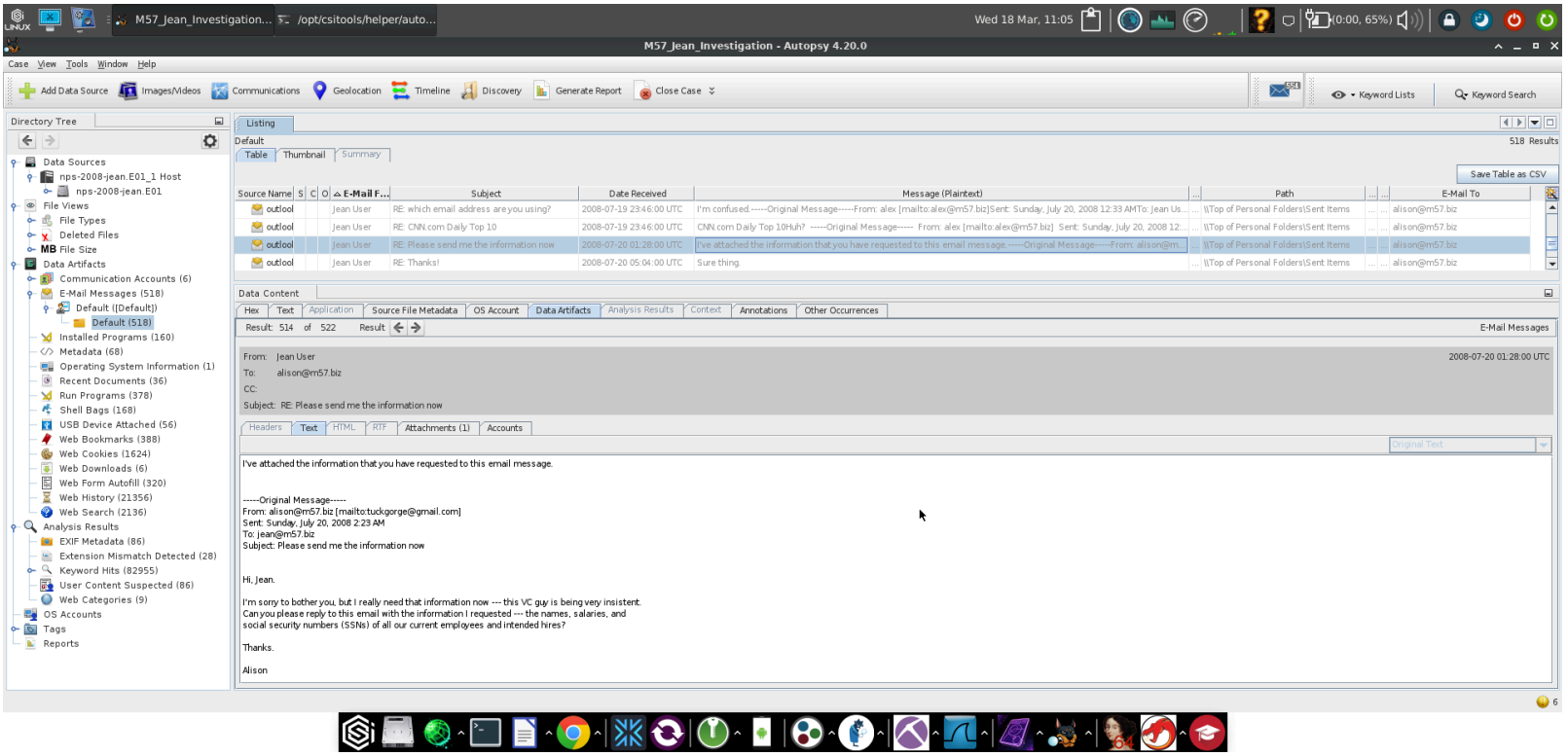


Figure 2: Email requesting confidential employee data including salaries and Social Security Numbers

5.3 Transmission of the Data via Email

Jean replied to the request stating that the information was attached. The attachment corresponds to the spreadsheet m57biz.xls, confirming that confidential data was transmitted externally.

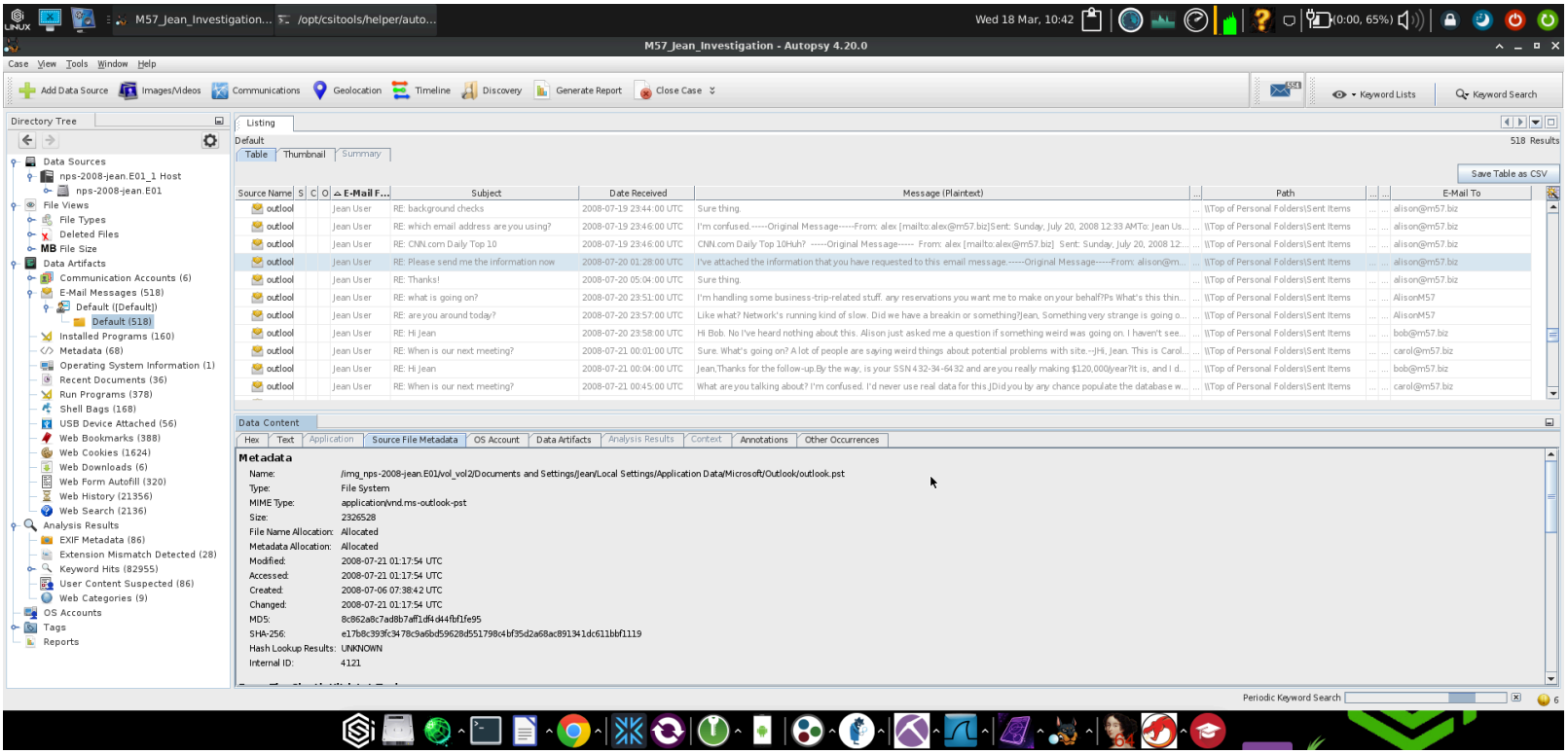


Figure 3: Email from Jean confirming transmission of the requested information as an attachment

5.4 Transmission to an External Address

Email header information shows communication with a non-corporate email address, confirming that the data was sent outside the organisation.

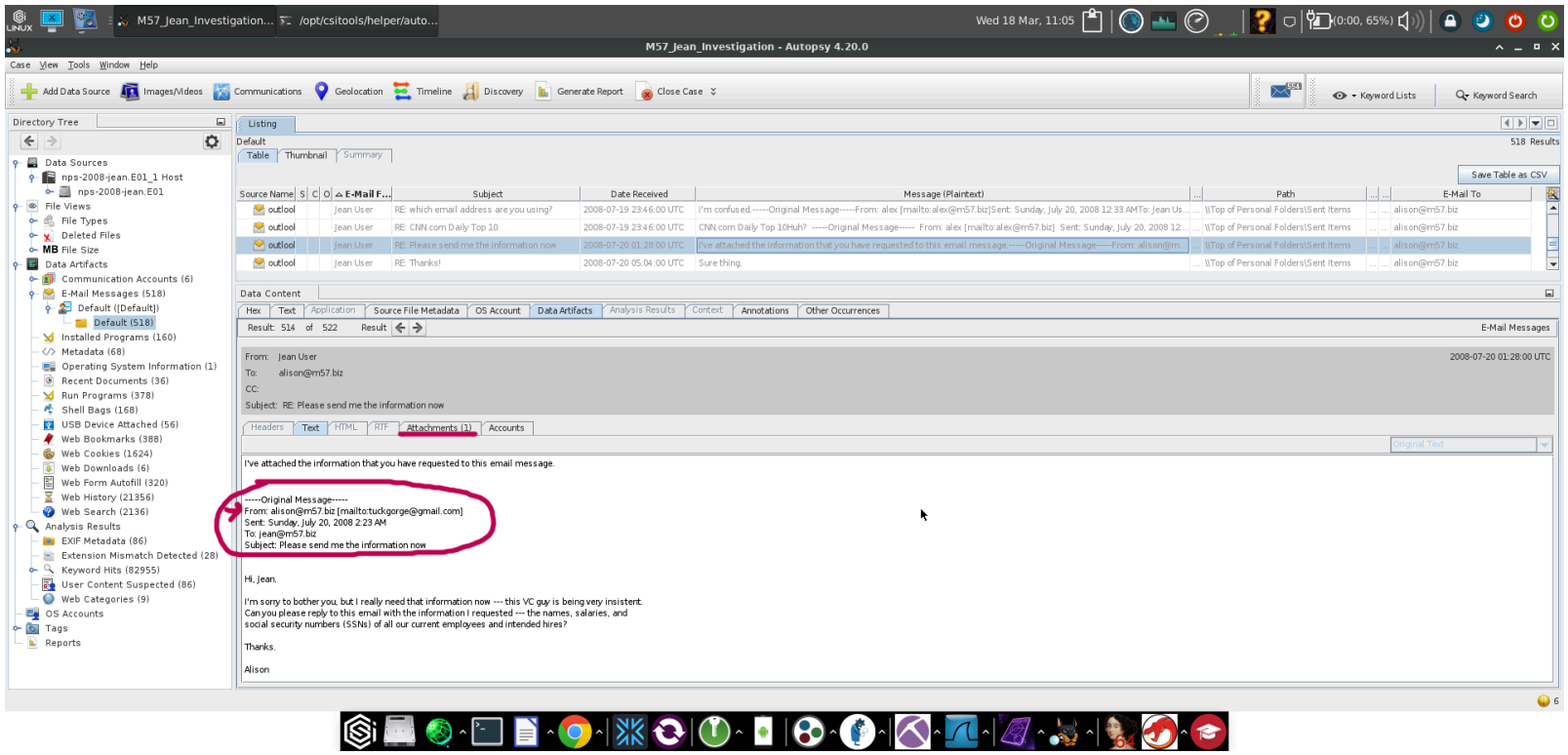


Figure 4: Email header showing communication with an external address

5.5 Evidence of Public Disclosure

Subsequent emails report that employees discovered their Social Security Numbers online, confirming public exposure of the transmitted data. Employees confirmed that the leaked details were accurate.

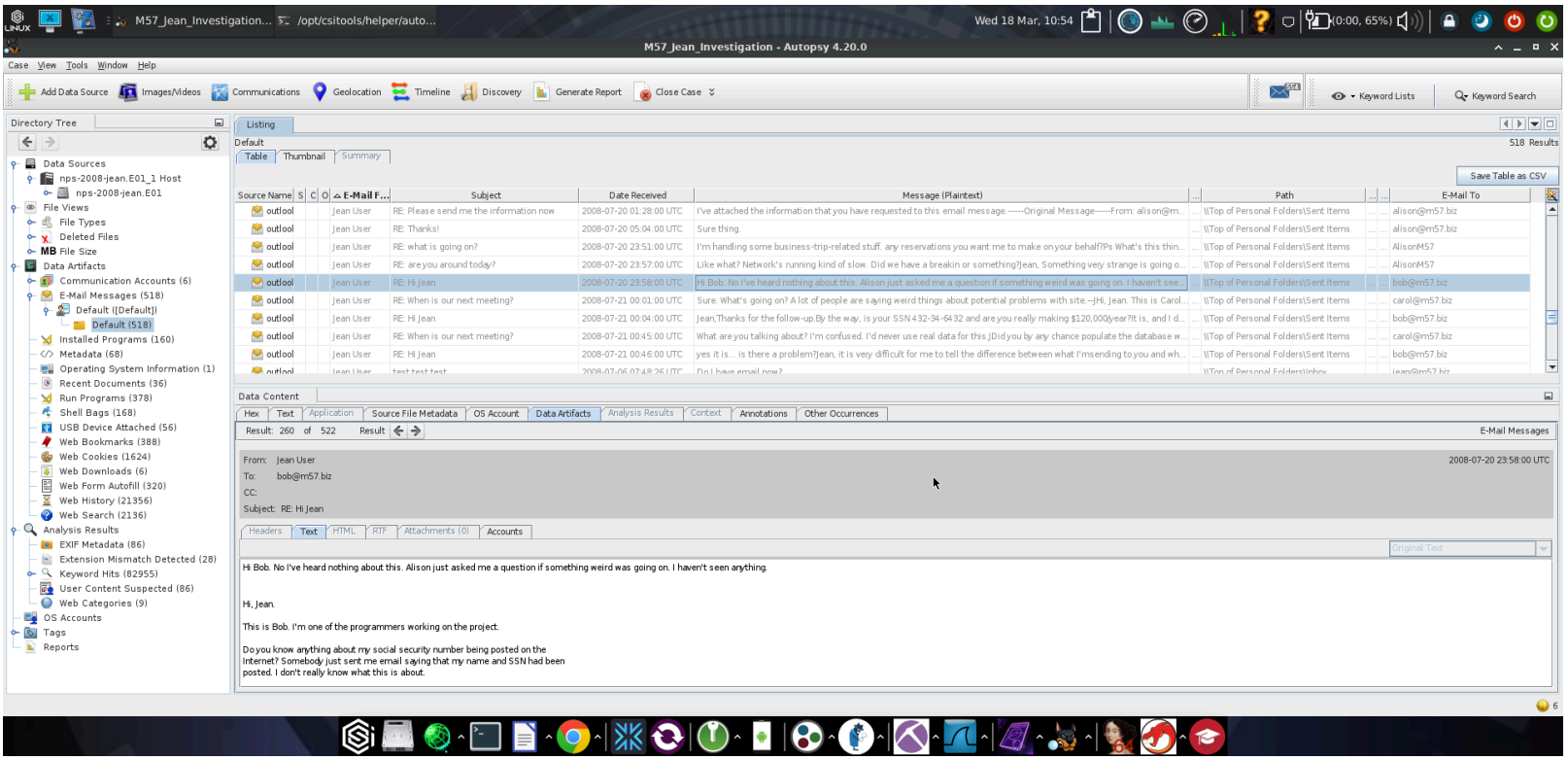


Figure 5: Email reporting public exposure of employee personal information

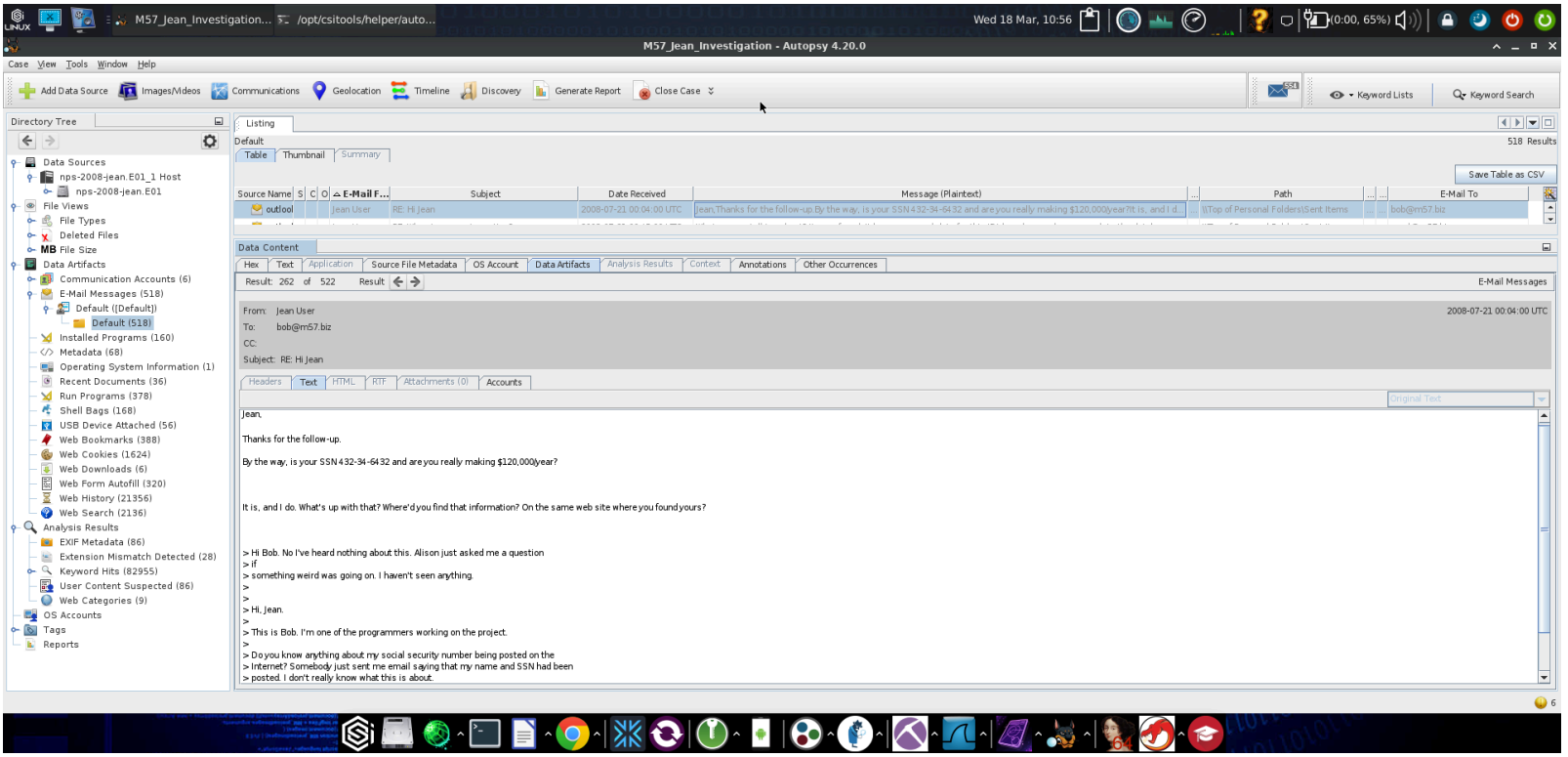


Figure 6: Email confirming accuracy of leaked SSN and salary data

5.6 Organisational Impact and Awareness

Internal emails indicate widespread concern and investigation into possible causes, including website or database issues.

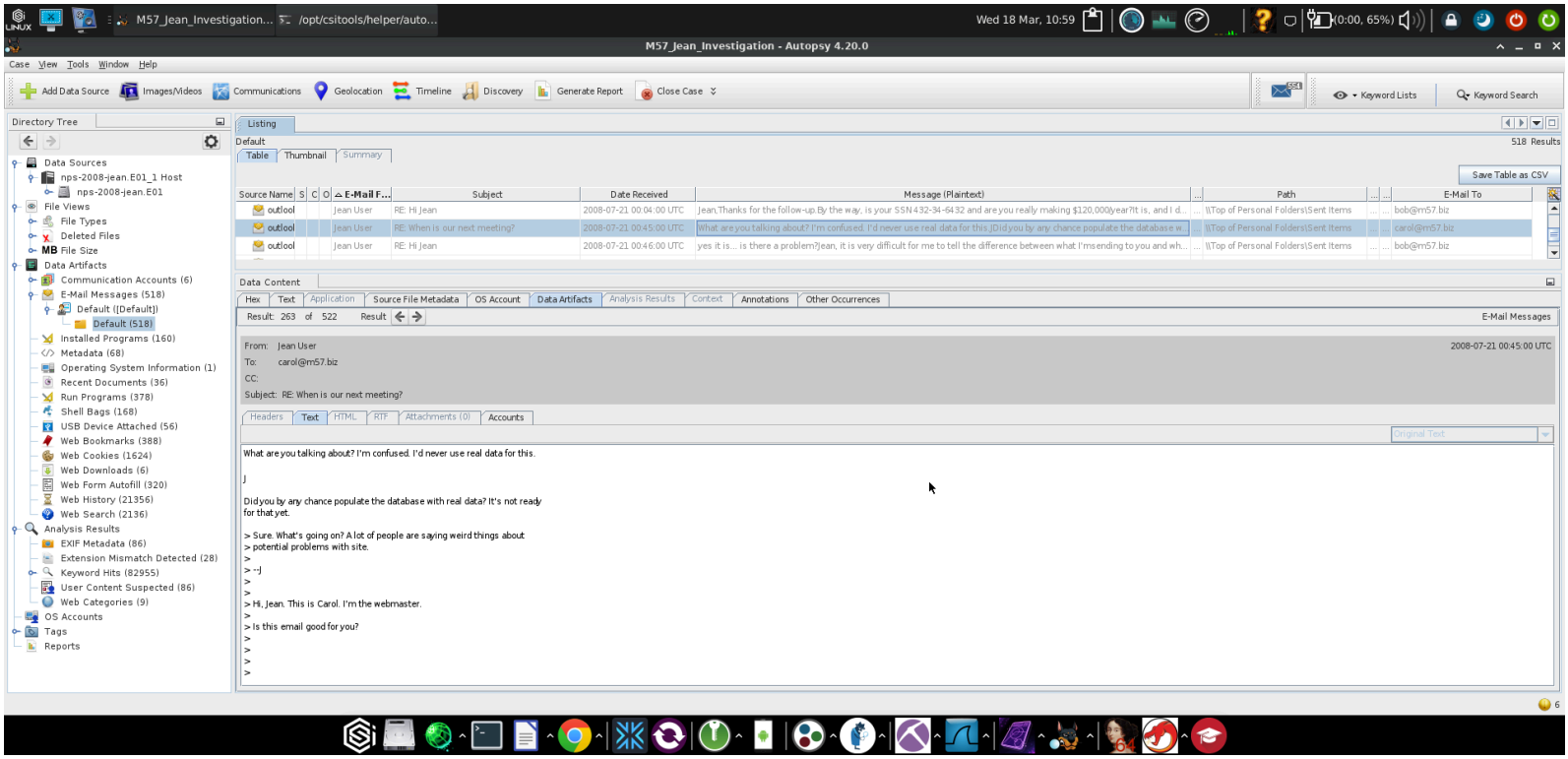


Figure 7: Email from webmaster discussing potential issues related to the website and data

5.7 Timeline Table

Date (2008)	Time (UTC)	Event Description	Supporting Evidence
July 20th	01:28	Jean sends email with attachment m57biz.xls to Alison	Sent email with attachment artefact
July 20th	02:23	Alison (m57.biz) forwards urgent request from external Gmail account	Email header shows mailtotuckgorge@gmail.com
July 20th	23:57	Alison emails Jean reporting suspicious activity	Internal email communication
July 20th	23:58	Bob emails Jean about SSN data appearing online	Staff report of exposure
July 21st	00:40	Jean replies denying knowledge of leak	Internal reply email
July 21st	00:45	Jean contacts Carole (webmaster) about situation	Email to Carole
July 21st	00:46	Follow-up email requesting confirmation of sensitive data	Internal communication
July 21st	01:17	Outlook PST modified during ongoing mailbox activity	PST metadata artefact

Figure 8: Timeline of Key Events in the Data Exfiltration Incident (UTC)

5.8 Summary of Involvement

There is enough evidence to suggest that Jean (sender), Alison (requesting), Bob, and other employee victims and Carol (the webmaster) were involved. However, there was no evidence of malicious or intentional wrongdoing on the part of internal staff in this incident. The incident appears to have occurred due to a social engineering attack that was able to exploit trust within the organisation. The fraudulent request made from Alison's email account suggests that either her email account was hacked or that a third party used an impersonation.

6. Supporting Evidence

Key evidence consisted of email communications extracted from Jean's Outlook PST archive. These emails document the request for confidential information, the transmission of an attachment containing employee data, and subsequent reports confirming public exposure. The presence of the spreadsheet attachment provides direct evidence of data exfiltration.

7. Investigative Leads, Assumptions and Limitations

During the course of the forensic examination, several investigative leads were identified alongside necessary assumptions and recognised limitations arising from the available evidence.

7.1 Investigative Leads

The primary investigative lead was an urgent email request for confidential employee data, including Social Security Numbers. The involvement of a non-corporate email address suggested that the information may have been transmitted outside organisational control. Reports from employees that their data appeared online further confirmed the occurrence of a breach.

7.2 Assumptions

Several assumptions were made in order to interpret the available evidence:

- It was assumed that the recovered disk image accurately represents the system state at the time of acquisition and has not been altered post seizure.
- It was assumed that timestamps contained within files and emails reflect the system clock accurately.

These assumptions are standard in forensic analysis when corroborating evidence supports their validity.

7.3 Limitations

The investigation was subject to several limitations that restrict the certainty of conclusions:

- Only a single host system was analysed such as no server side logs, network captures, or external systems were available for correlation.
- The identity of the external recipient could not be conclusively validated.

Additionally, deleted or encrypted data that may exist outside the recovered artefacts could not be examined within the scope of this investigation.

8. Recommendations

All companies should be implementing strict regulations on the transfer/sharing/distribution of sensitive data, including encrypted communication only to pre-authorized employees through pre-approved channels, in addition, employees need training to identify social engineering and phishing attempts. Verification protocols are also necessary for any type of urgency or non-routine request related to confidential data. Data Loss Prevention (DLP) solutions can be utilized to monitor and eliminate unauthorized data transfers.

9. References

Carrier, B., 2005. *File System Forensic Analysis*. Boston: Addison-Wesley.

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